

Program – 7: Deploy a Web Application to Kubernetes using Minikube

This experiment demonstrates deploying an Nginx web server on a Kubernetes cluster using a Deployment and exposing it using a Service in Minikube.

Prerequisites

- Minikube installed

- kubectl installed

Procedure

Step 1: Start Minikube

`minikube start`

Step 2: Check Minikube Status

`minikube status`

Step 3: Create Deployment (nginx-deployment.yaml)

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
name: nginx-deployment
```

```
spec:
```

```
replicas: 2
```

```
selector:
```

```
matchLabels:
```

```
app: nginx
```

```
template:
```

```
metadata:
```

```
labels:
```

```
app: nginx
```

```
spec:
```

```
containers:
```

```
- name: nginx
```

```
image: nginx:latest
```

```
ports:- containerPort: 80
```

Step 4: Apply Deployment

`kubectl apply -f nginx-deployment.yaml`

Step 5: Verify Deployment and Pods

`kubectl get deployments`

`kubectl get pods`

Step 6: Expose Deployment as a Service

`kubectl expose deployment nginx-deployment --type=NodePort --port=80`

Step 7: Access the Web Application

`minikube service nginx-deployment`

This opens the Nginx web page in your browser.

Step 8: Clean Up Resources

```
kubectl delete pods --all
kubectl delete svc --all
kubectl delete deployments --all
```

Expected Output

Nginx default welcome page displayed in the browser.

```

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
📦 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
rv24mc074_poornimat@lenovo:~/program7$ minikube service nginx-deployment.yaml

❌ Exiting due to SVC_NOT_FOUND: Service 'nginx-deployment.yaml' was not found in 'default' namespace.
You may select another namespace by using 'minikube service nginx-deployment.yaml -n <namespace>'. Or list out all the s
rvices using 'minikube service list'

rv24mc074_poornimat@lenovo:~/program7$ minikube service nginx-deployment

❌ Exiting due to SVC_NOT_FOUND: Service 'nginx-deployment' was not found in 'default' namespace.
You may select another namespace by using 'minikube service nginx-deployment -n <namespace>'. Or list out all the servic
s using 'minikube service list'

rv24mc074_poornimat@lenovo:~/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
rv24mc074_poornimat@lenovo:~/program7$ kubectl get svc
AME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
ubernetes    ClusterIP     10.96.0.1     <none>         443/TCP          5d
ginx-servicee NodePort      10.111.8.206  <none>         80:30659/TCP     5d
rv24mc074_poornimat@lenovo:~/program7$ minikube service nginx-servicee
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-servicee	80	http://192.168.49.2:30659

```

🌐 Opening service default/nginx-servicee in default browser...
rv24mc074_poornimat@lenovo:~/program7$ minikube service nginx-servicee
|
```

← → 🔄 ⚠ Not secure 192.168.49.2:30659 ☆ 📦 Finish update ⋮

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.